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NPTEL ONLINE CERTIFICATION COURSE

Charging Infrastructure

Lecture-38

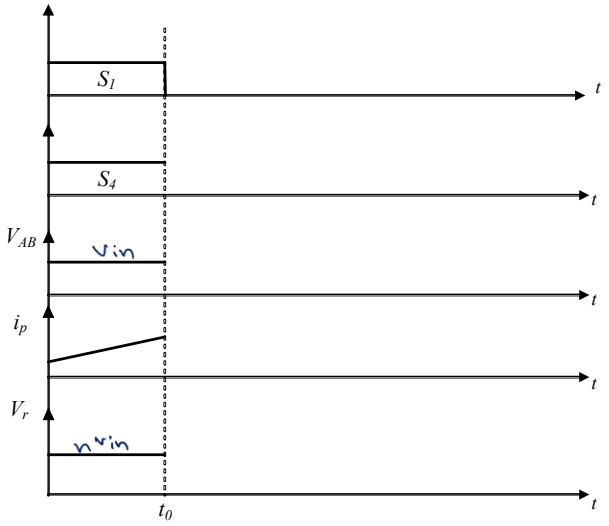
Phase Shifted Full Bridge (PSFB) Converter-II

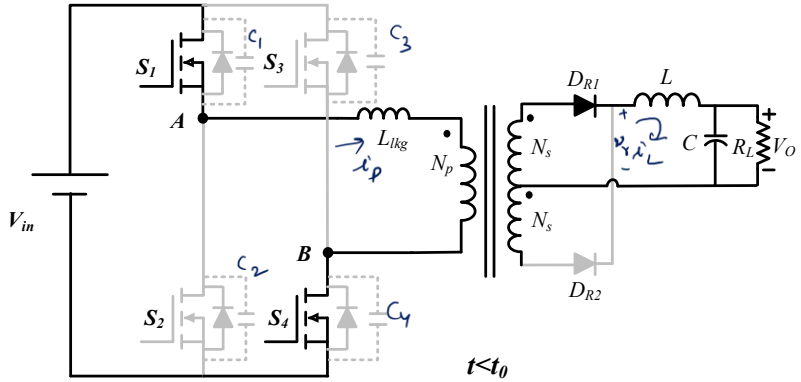
Dr. Apurv Kumar Yadav

Department of Electrical Engineering



PSFB Operating Modes (Mode-0)





S_1 & S_4 is conducting

$v_{C4} = 0$

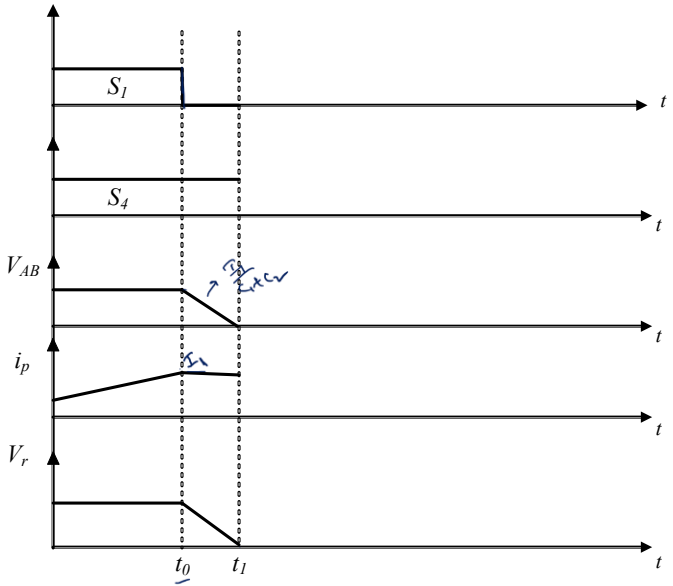
$v_{C3} = v_{in}$

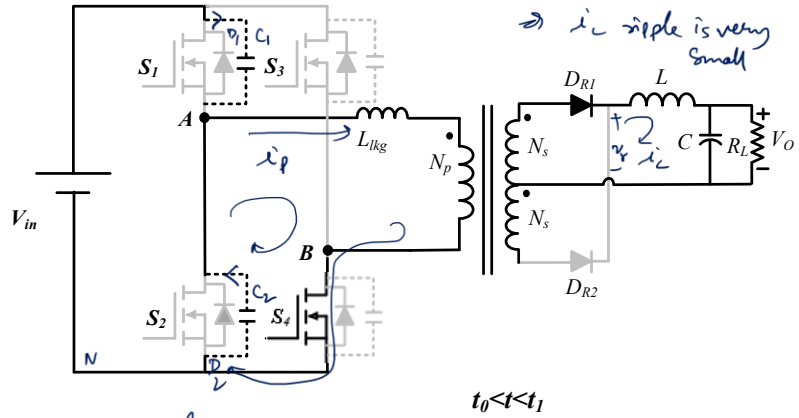
$v_{C1} = 0$

$v_{C2} = v_{in}$

Source: Xinbo Ruan, "Soft-Switching PWM Full-Bridge Converters: Topologies, Control, and Design", 1st Ed. Wiley, 2014

PSFB Operating Modes (Mode-1)





$t_0 < t < t_1$

L is large
⇒ i_L ripple is very small

C_2 is discharging from $V_{in} \rightarrow 0$
⇒ $v_{C_2} \downarrow$

C_1 is charging from 0
⇒ $v_{C_1} \uparrow$

$v_{AB} = v_{C_2}$

$i_{C_1} + i_{C_2} = i_p$

$\Rightarrow \frac{dv}{dt} = \frac{I_1}{C_1 + C_2}$

Source: Xinbo Ruan, "Soft-Switching PWM Full-Bridge Converters: Topologies, Control, and Design", 1st Ed. Wiley, 2014

PSFB Operating Modes (Mode-1) Cont.

$i_p(t_0) \simeq i_p(t_1) \simeq I_1$

$v_{c_1}(t) = \frac{I_1}{C_1 + C_2}(t - t_0)$ $\quad (C_1 \simeq C_2 \simeq C)$

$\Rightarrow v_{c_1}(t) = \frac{I_1}{2C}(t - t_0)$

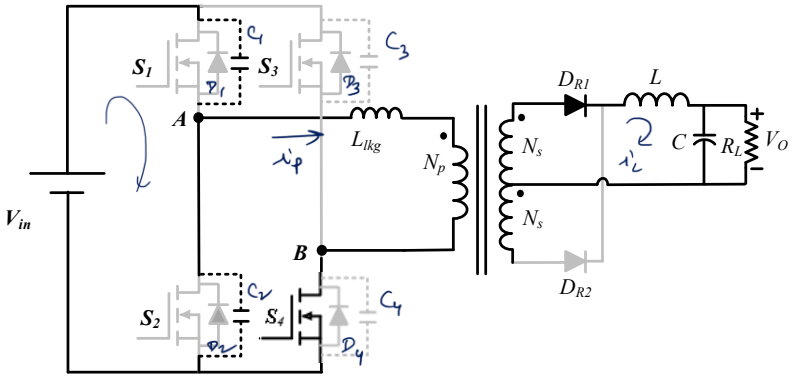
$\Rightarrow v_{c_2}(t) = V_{in} - \frac{I_1}{2C}(t - t_0)$

at $t = t_1$ $\Rightarrow v_{c_1}(t_1) = V_{in}$

$\Rightarrow v_{c_2}(t_1) = 0$

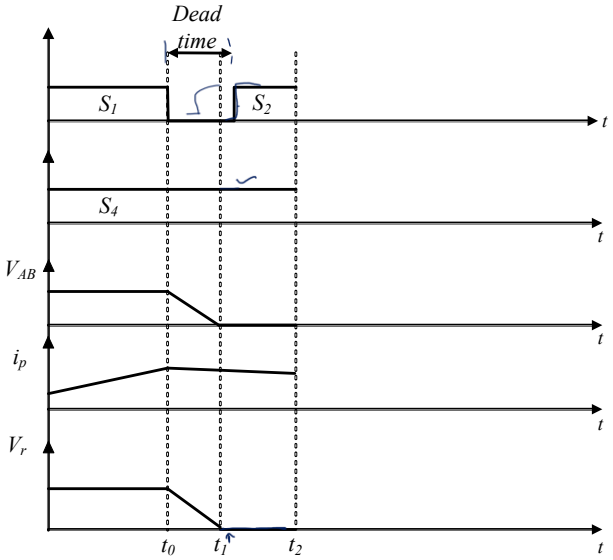
$\Rightarrow$ $t_{01} = t_1 - t_0 = \frac{2C\,V_{in}}{I_1}$

after t_{01} $\rightarrow$ the $v_{c_2} = 0$ $\Rightarrow$ the diode can turn-on at $t = t_1$

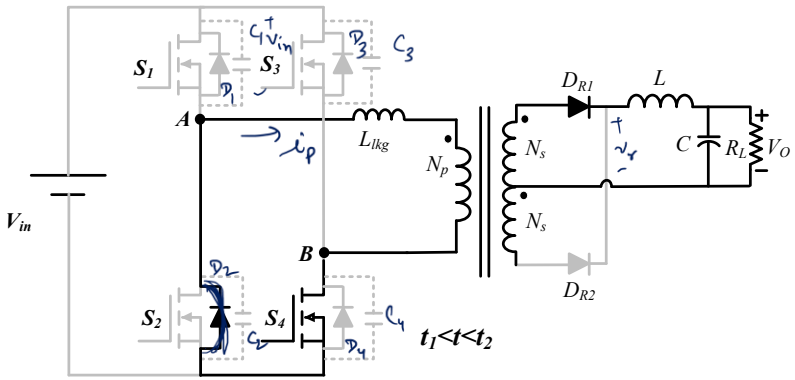


Source: Xinbo Ruan, "Soft-Switching PWM Full-Bridge Converters: Topologies, Control, and Design", 1st Ed. Wiley, 2014

PSFB Operating Modes (Mode-2)



$v_{C_1} = v_{in}$



⇒ Since the $v_{C_2} = 0$

⇒ the diode D_2 turn-on

$i_{D_2} = i_p$

Since D_2 is on, after some time the switch S_2 can turn-on with ZVS

Source: Xinbo Ruan, "Soft-Switching PWM Full-Bridge Converters: Topologies, Control, and Design", 1st Ed. Wiley, 2014

PSFB Operating Modes (Mode-2)

if, Deadtime $> t_{01}$ ⟶

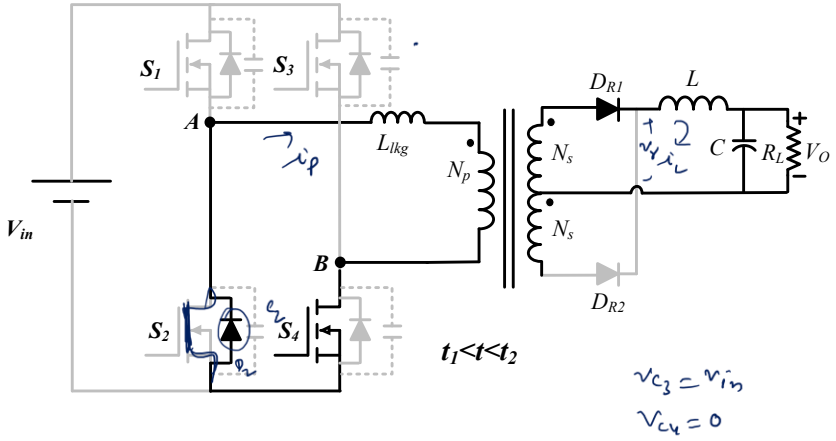
$\simeq$ Deadtime $> \frac{2CV_{in}}{I_1}$ $\Rightarrow$ ZVS turn on of S_2 switch is possible

$\Rightarrow$ after sometime between t_1 & t_2 the gate pulse to S_2 can be given, which ensure ZVS turn-on of S_2 switch (leading leg switch)

to ensure ZVS, $\Rightarrow$ Deadtime (leading leg) $> \frac{2CV_{in}}{I_1}$ $\left(C \simeq C_1 \simeq C_2\right)$

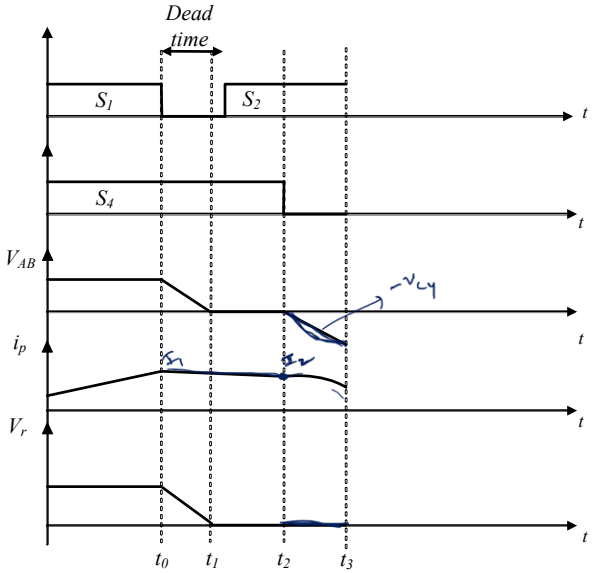
$i_L \simeq I_o$

$\Rightarrow$ $i_p \simeq n i_L$ $\Rightarrow$ $I_1 \simeq n I_o$



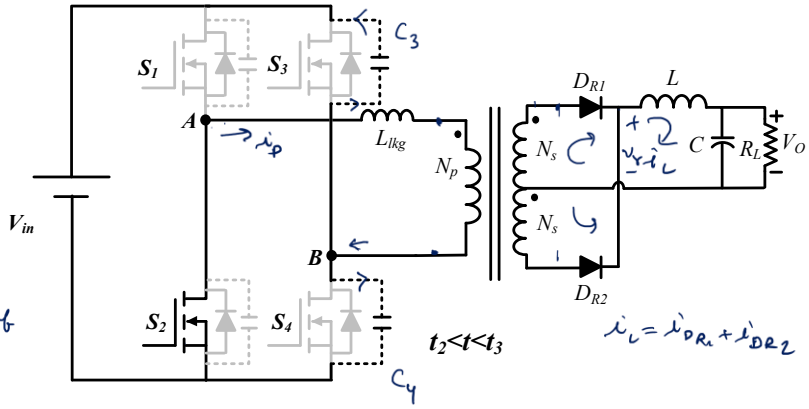
Source: Xinbo Ruan, "Soft-Switching PWM Full-Bridge Converters: Topologies, Control, and Design", 1st Ed. Wiley, 2014

PSFB Operating Modes (Mode-3)



$t_2 - t_0$ = Phase-shift

at $t = t_2$ the Switch S_4 → turns-off (lagging leg Switch)



C_3 is discharging from V_{in}

C_4 is charging from 0V

⇒ $v_{AB} = -v_{C4}$ (v_{C4} ↑)

⇒ i_p will no longer be reflected i_L current

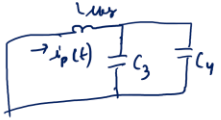
⇒ D_{R1} & D_{R2} gets forward biased ⇒ $v_r = 0$

Source: Xinbo Ruan, "Soft-Switching PWM Full-Bridge Converters: Topologies, Control, and Design", 1st Ed. Wiley, 2014

PSFB Operating Modes (Mode-3)

V_{AB} is imposed upon L_{lkg}

L_{lkg} is in resonance with $(C_3 + C_4)$



$$i_p(t) = I_2 \cos(\omega_1 (t - t_2))$$

$$v_{C_4}(t) = \frac{1}{2C} \int i_p(t) \cdot dt$$

$$= \frac{1}{2C\omega_1} I_2 \sin(\omega_1 (t - t_2))$$

$$v_{C_4}(t) = \sqrt{\frac{L_{lkg}}{2C}} \cdot I_2 \sin(\omega_1 (t - t_2))$$

$$\Rightarrow v_{C_3}(t) = v_{in} - v_{C_4}(t)$$

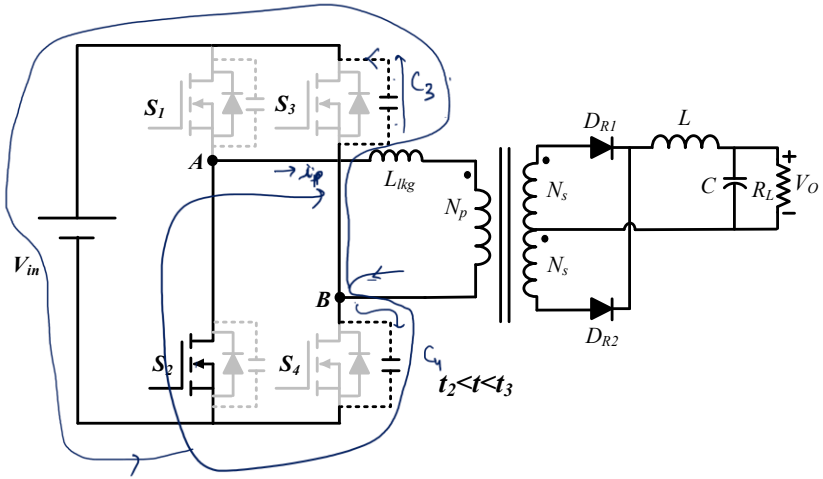
$$= v_{in} - \sqrt{\frac{L_{lkg}}{2C}} \cdot I_2 \sin(\omega_1 (t - t_2))$$

ω_1 → resonance freq.

$$\omega_1 = \frac{1}{\sqrt{L_{lkg}(C_3 + C_4)}}$$

$$\Rightarrow C_3 \simeq C_4 = C$$

$$\Rightarrow \omega_1 = \frac{1}{\sqrt{2 L_{lkg} C}}$$



PSFB Operating Modes (Mode-3)

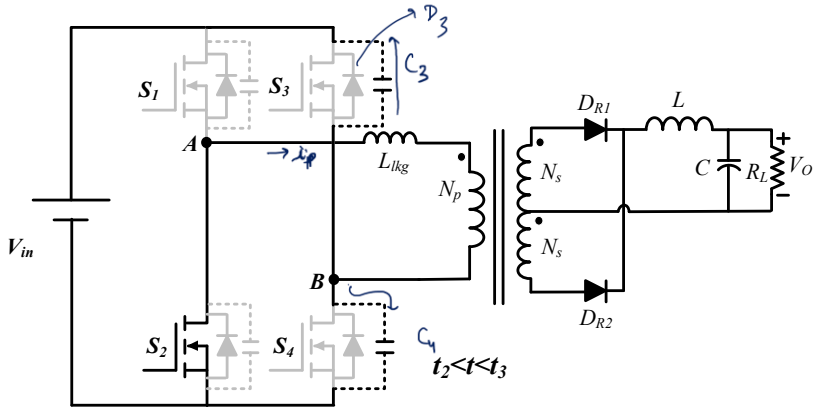
at $t = t_3$

$t_{23} = t_3 - t_2$

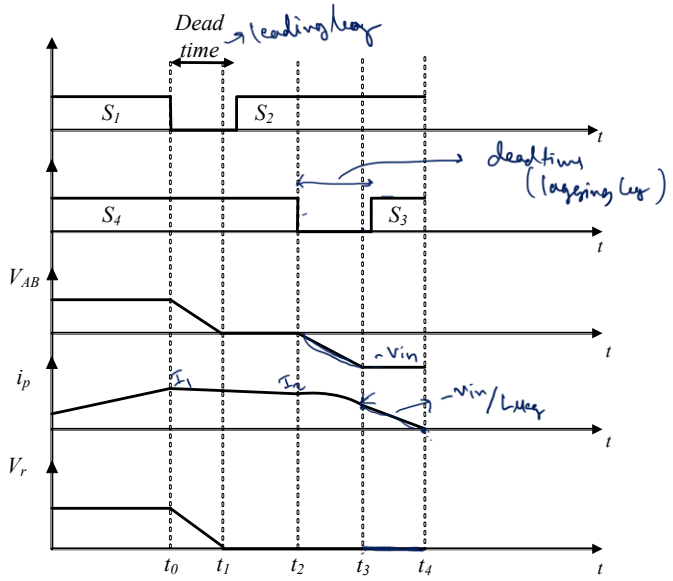
$\Rightarrow$ i.e. $v_{C3} = 0 \Rightarrow v_{C4} = v_{in}$

$\Rightarrow$

$\Rightarrow$ $t_{23} = \frac{1}{\omega_1}\left(\sin^{-1}\frac{v_{in}}{I_2}\sqrt{\frac{2C}{L_{lkg}}}\right)$



PSFB Operating Modes (Mode-4)

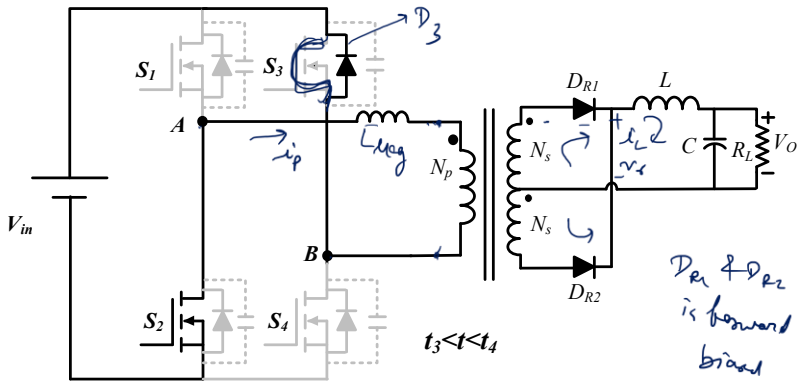


deadtime > t_{23} (lagging leg)

$\Rightarrow$ deadtime (lagging leg) > $\frac{1}{\omega_1}\left(\sin^{-1}\frac{V_{in}}{I_2}\sqrt{\frac{2C}{L_{lag}}}\right)$

$\Rightarrow$ $v_{c_3}(t)$ goes to '0'

$\Rightarrow$ Diode D_3 turns on



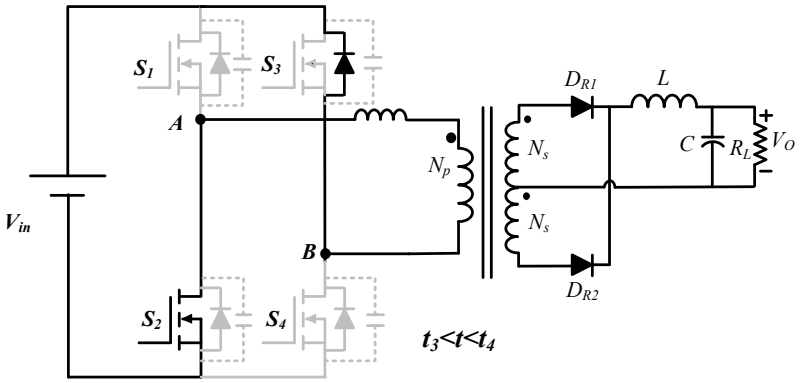
Source: Xinbo Ruan, "Soft-Switching PWM Full-Bridge Converters: Topologies, Control, and Design", 1st Ed. Wiley, 2014

PSFB Operating Modes (Mode-4)

$$i_p(t) = I_p(t_3) - \frac{V_{in}}{L_{lkg}}(t - t_3)$$

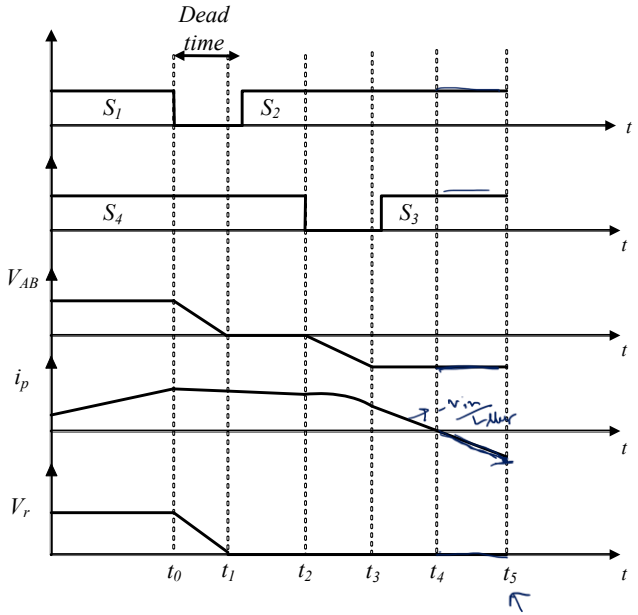
at $t = t_4$, the $i_p(t_4) = 0$

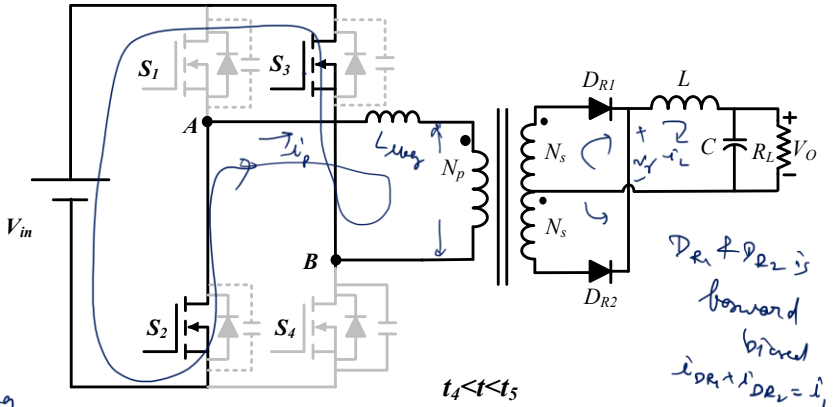
$$\Rightarrow \quad t_{43} = t_4 - t_3 = L_{lkg}\frac{I_p(t_3)}{V_{in}}$$



Source: Xinbo Ruan, "Soft-Switching PWM Full-Bridge Converters: Topologies, Control, and Design", 1st Ed. Wiley, 2014

PSFB Operating Modes (Mode-5)





$v_{AB} \simeq -v_{in}$ is coming across L_{leg}

⇒ winding $v_{lg} = 0$

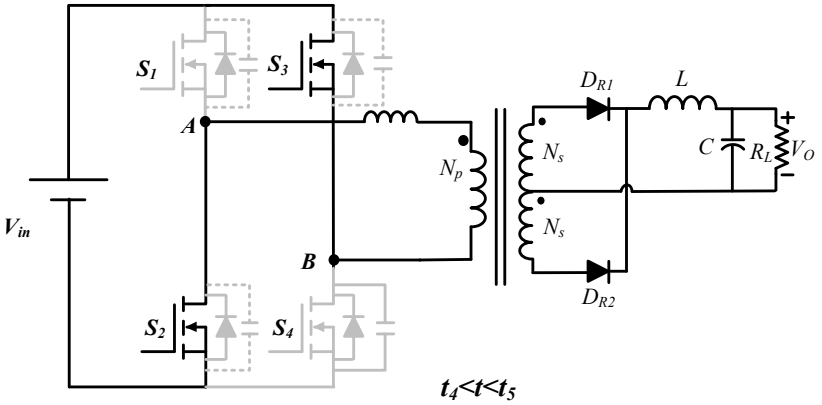
$$i_p(t) = \frac{-v_{in}}{L_{leg}}(t - t_4)$$

⇒ at $t = t_5$, $i_p(t_5) = -n\, i_L \simeq -n I_o$

Source: Xinbo Ruan, "Soft-Switching PWM Full-Bridge Converters: Topologies, Control, and Design", 1st Ed. Wiley, 2014

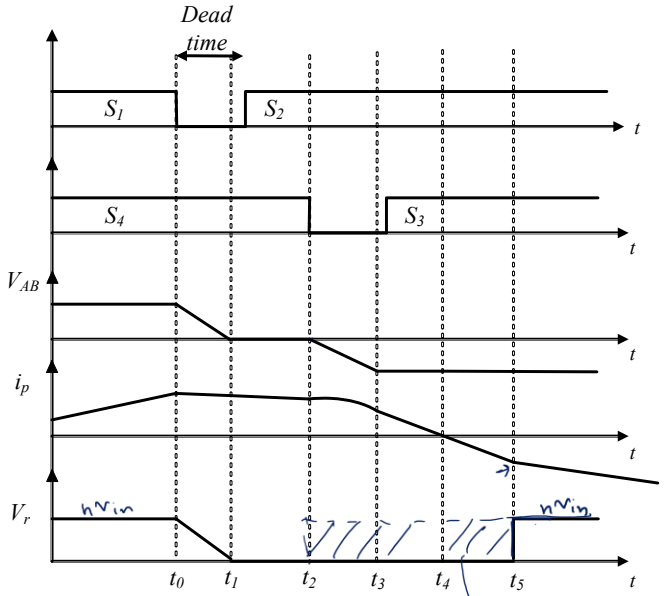
PSFB Operating Modes (Mode-5)

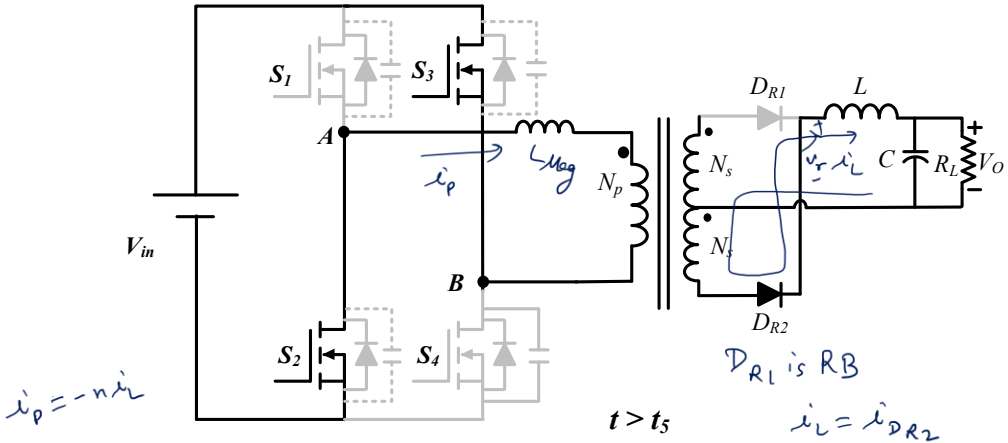
$t_{43} = t_5 - t_4 = \frac{L_{lkg}\, n I_o}{V_{in}}$



Source: Xinbo Ruan, "Soft-Switching PWM Full-Bridge Converters: Topologies, Control, and Design", 1st Ed. Wiley, 2014

PSFB Operating Modes (Mode-6)





$V_o = n V_{in} \cdot D$

$\Rightarrow V_o = n V_{in} (D - D_{loss})$

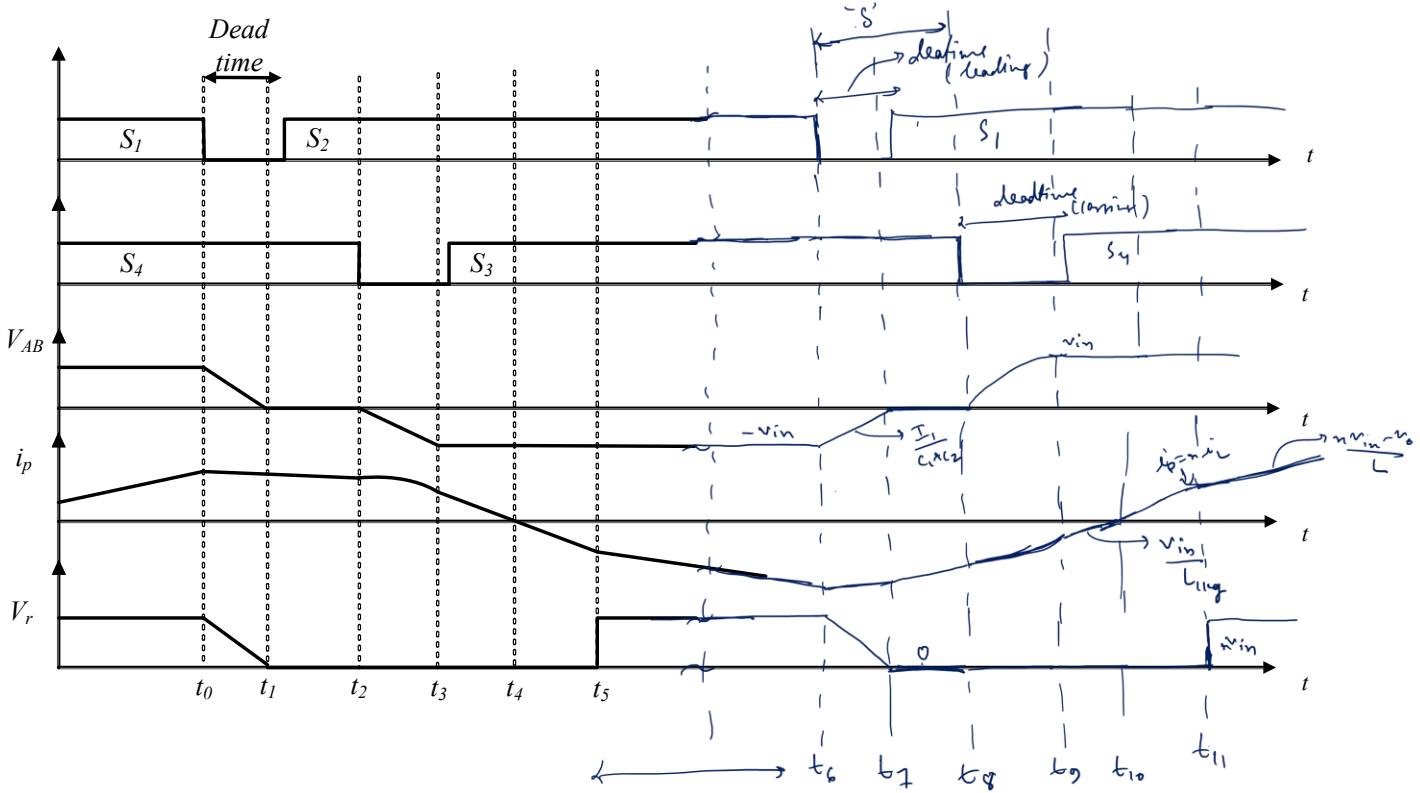
D_{loss} → corresponds to time t_2 to t_5

Source: Xinbo Ruan, "Soft-Switching PWM Full-Bridge Converters: Topologies, Control, and Design", 1st Ed. Wiley, 2014

PSFB Operating Modes

t_5 to t_6 is very large

$t_0 - t_5$ are small durations as compared to T_s



Source: Xinbo Ruan, "Soft-Switching PWM Full-Bridge Converters: Topologies, Control, and Design", 1st Ed. Wiley, 2014

Summary

Mode-0

S_1, S_4, D_{R1}

Mode-1

C_1, C_2, D_{R1}, S_4

Mode-2

$D_2, S_4, D_{R1} \rightarrow S_2, S_4, D_{R1}$

Mode-3

$S_2, C_3, C_4, D_{R1}, D_{R2}$

Mode-4

$S_2, D_3, D_{R1}, D_{R2} \rightarrow S_2, S_3, D_{R1}, D_{R2}$

Mode-5

S_2, S_3, D_{R1}, D_{R2}

mode-6

S_2, S_3, D_{R2}

$t_2 - t_5$ are the duty loss duration

required $\rightarrow t_{deadtime(leading)} > t_{01}$

$> \frac{2C V_{in}}{I_1}$

required $\rightarrow t_{deadtime(lagging)} > t_{23}$

$> \frac{1}{\omega_1} \sin^{-1}\left(\frac{V_{in}}{I_2}\sqrt{\frac{2C}{L_{lkg}}}\right)$

Source: Xinbo Ruan, "Soft-Switching PWM Full-Bridge Converters: Topologies, Control, and Design", 1st Ed. Wiley, 2014

necessary condition for ZVS of leading leg (turn-on)

$$\frac{1}{2}\left(L_{Mag} + \frac{L}{n^2}\right) I_1^2 > \frac{1}{2} C_1 V_{in}^2 + \frac{1}{2} C_2 V_{in}^2$$

necessary condition for ZVS of lagging leg (turn-on)

$$\frac{1}{2}\left(L_{Mag}\right) \cdot i_p^2 > \frac{1}{2} C_3 V_{in}^2 + \frac{1}{2} C_4 V_{in}^2$$

→ the lagging leg has problem in achieving ZVS during turn-on, when we have light load, small L_{Mag}, large C_3 & C_4, large input voltage

$$\text{Duty loss} = D_{loss} = \frac{t_5 - t_2}{T_s/2} = \frac{t_{23} + t_{43} + t_{54}}{T_s/2}$$

t_{23} ≃ small values as compare to t_{43} & t_{54}

Source: Xinbo Ruan, "Soft-Switching PWM Full-Bridge Converters: Topologies, Control, and Design", 1st Ed. Wiley, 2014

$t_{2-5} = t_{43} + t_{54}$

$= \frac{L_{Lkg}\, i_p(t_3)}{V_{in}} + \frac{L_{Lkg}\, i_p(t_5)}{V_{in}}$

$= \frac{L_{Lkg}}{V_{in}} \left[I_2 + n I_L(t_5) \right]$

$I_1 \simeq I_2 \simeq n I_o$

$I_L(t_5) \simeq I_o$

$t_{2-5} = \frac{L_{Lkg}}{V_{in}} \left[2n I_o \right]$

$\Rightarrow D_{loss} = \frac{t_{2-5}}{T_{s/2}} = \frac{\frac{L_{Lkg}(2nI_o)}{V_{in}}}{T_{s/2}}$ $\left[T_s = \frac{1}{f_{sw}} \right)$

$\Rightarrow$ $\boxed{D_{loss} = \frac{4 L_{Lkg} n I_o \cdot f_{sw}}{V_{in}}}$

$\Rightarrow$ to achieve $\rightarrow L_{Lkg}\uparrow \rightarrow D_{loss}\uparrow$

$\Rightarrow$ $V_{in}\uparrow \rightarrow D_{loss}\downarrow$; $I_o\uparrow \rightarrow D_{loss}\uparrow$; $f_{sw}\uparrow \rightarrow D_{loss}\uparrow$

$\Rightarrow$ In one wants to have ZVS of both the legs, they have compromise on D_{loss}

$\Rightarrow V_o = n V_{in} (D - D_{loss})$

Thank You